UNIVERSITY OF CAPE COAST

DEPARTMENT OF PHYSICS

PHY 101 (GENERAL PHYSICS)

TUTORIAL SET ONE

1. What do you mean by physical quantity?

2. Differentiate between fundamental and derived unit

3. Write the definition of Dimensions.

4. Write the dimensional formula of the following physical quantities

(i) Potential Energy (ii) Power (iii) Surface Tension (iv) Strain (v) stress

5. Add the following velocities and leave your answer in two significant figure

0.0234 m/s, 1.234 m/s, 0.00032 m/s, 1.0001 m/s and 0.1011 m/s

6. Check the correctness of the relation λ = h /mv; where λ is wavelength, h- Planck's constant, m is mass of the particle and v - velocity of the particle.

7. Check the correctness of the following relation by using method of dimensions

(i) v = u + at

(ii) $F = \frac{MV}{r^2}$

(iii) $v^2 - u^2 = 2as$

8. Convert velocity of 100 km/hr into m/s

9. what is the formula, S.I units and physical quantity of the following

(a) $[ML^2T^{-3}]$

(b) $[M^1L^2T^{-2}]$

10. The velocity of a particle is given in terms of time t by the equation v = At + B/t + C. What are the dimensions of A, B and C?

11. The speed of sound in a gas is given by $V = k\rho^x P^y$, where V=speed, Pressure, ρ is density

and k is dimensionless constant. Find the constants x and y and write an equation relating V, and ρ

12. The frequency *f* of a transverse wave in a stretched string depends on the length L, tension T, and linear density *m*. Deduce the formula for f in terms of L, T and m using dimensional analysis.

13. Newton's law of universal gravitation is represented by

$$F = \frac{GMm}{r^2}$$

Here *F* is the magnitude of the gravitational force exerted by one small object on another, *M* and *m* are the masses of the objects, and *r* is a distance. Force has the S.I units kg/ms^2. What are the SI units of the proportionality constant *G*?

14. How many significant figures do have in the following?

(i) 0.24560 cm

(ii) 0.000023120 kg

15. On an interstate highway in a rural region of Wyoming, a car is traveling at a speed of 38.0 0 m/s. Is this car exceeding the speed limit of 75.0 mi/h? What if the driver is from outside U.S. and is familiar with speeds measured in km/h? What is the speed of the car in km/h?

16. A force of 3N acts on 90" to a force of 4N. Find the magnitude and direction of the resultant, R.

17. A particle which is moving due east at 4ms" changes direction and starts to move due south at 3ms'. Find the change in velocity.

18. Consider two vectors A = 3i – 2j and B = – 4i + 3j. Calculate:

(i) 3B – A (ii) the magnitude of (i) (iii) the unit vector of (i)

19. A velocity of a particle moving at an angle of 120° to the horizontal is 24 km/hr. find the horizontal and vertical components of the velocity.

20. (a) what are the uses of dimensions?

(b) what are the limitations of dimensional analysis?